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Height Restrictions for Dual-energy X-ray Absorptiometry: What Are Our Options for Body Composition and Bone Health Precision?

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Abstract

Redinger, AL, Russell, JL, Allen, SMF, and Baker, BS. Height restrictions for dual-energy x-ray absorptiometry: what are our options for body composition and bone health precision? *J Strength Cond Res* XX(X): 000–000, 2024—Dual-energy X-ray absorptiometry (DXA) is commonly used for testing athlete's body composition, but many athletes are too tall. The first aim was to assess the viability of combining upper- and lower-body regions of interest (ROI), creating a combined scan for tall athletes who do not fit on the table and second, to provide practical solutions to DXA users who routinely scan tall athletes. Sixty subjects (34.8 ± 11.9 years; 171 ± 9 cm) completed 2 total-body DXA scans for baseline precision testing, using GE Lunar Prodigy (LP) or Hologic Horizon A (HA) models. Next, an upper body ROI from the skull to the distal femoral condyles was combined with a flipped scan (feet-to-head) ROI encompassing the proximal tibial plateau to the distal foot. Soft and bone tissue coefficient of variance (CV%) were calculated between the baseline scans and for the newly combined ROI scan. The combined ROI scan added 0.25–0.63% and 0.01–4.35% error rates for the LP and HA, respectively. An exploratory assessment of a GE Lunar iDXA demonstrated results similar to the HA with 4%+ error. The combined ROI scan is a user-friendly and precise method for older LP systems adding less than 1% error; however, newer DXA systems cannot use the stitched scan technique. Coaches and practitioners who use newer DXAs must prioritize consistently using the same boney landmarks (head, jaw, or feet) and ROI heights to provide precise longitudinal assessments of tall athletes' bone and body composition, until larger DXA tables become available.

Key Words: tall athletes, bone mineral content, musculoskeletal health, lean mass, fat mass

Introduction

The late 1980s saw the arrival of the first commercially available Dual-energy X-ray Absorptiometry (DXA) machine released by Hologic (Hologic, Bedford, MA). Today, DXA has become the primary clinical tool for osteoporosis assessments and the gold-standard research tool for body composition (2,16,23). Demand for DXA technology continues to grow as manufacturers improve image quality, accuracy, and precision, while adding new diagnostic features (16,22). One area that has seen a great increase in the usage of DXA technology is competitive athletics (8,25). Athlete populations engage in repetitive, sport-specific tasks over long durations and consecutive seasons, warranting serial assessments to determine musculoskeletal health and body composition adaptations. Investigations of this nature have become more prevalent with taller athletes competing in sports such as basketball, football, and volleyball (6,18). The combination of unique loading profiles and musculoskeletal overuse injury risk further amplifies the need for temporal evaluation within these cohorts. DXA precision is especially important for serial testing and is often expressed as the coefficient of variation (CV%) or the least significant change which findings must exceed to be considered relevant (3,7,11). When comparing serial scans, it is vital that the DXA CV% is small, allowing for confidence in the technicians' repeatability to observe changes in bone, adipose tissue, and lean mass over time.

Although body fat percentage has traditionally been emphasized in reporting, musculoskeletal health is becoming more commonly discussed and serves as an important factor for competitive longevity (21,24). Aspects such as lean mass, bone mineral distribution, and tissue asymmetries can act as pertinent tissue indicators for performance outcomes and musculoskeletal injury risk (14,20). Seasonal evaluations should serve as a focal point for athletes, developmental personnel, and medical staff because of the dynamic impact of training and performance across all tissues (26). Furthermore, site-specific assessments can be used to help inform users of isolated tissue asymmetries, further identifying injury risk (4,10,13). Because of this practical and clinical importance, it is essential for practitioners to have standardized and time-effective scan and analysis protocols, ensuring accurate and precise DXA data evaluations across seasons.

A key limitation to using DXA with exceedingly tall athletes is the table size. Although DXA table sizes have increased in recent years, most DXA machines cannot accommodate an athlete who exceeds 193 cm (12). To circumnavigate this limitation, past procedures have encouraged technicians to exclude skeletal segments such as portions of the skull or shank (17). For instance, Nana et al. (17) suggested positioning in which a portion of the head may be omitted to include the entirety of the rest of the body. However, in very tall athletes, such as those competing in professional basketball leagues, many individuals are so tall that omission of the head will still not result in the full inclusion of the

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feet. In addition, recent work from Hew-Butler et al. (15) indicated that inclusion of the head is pertinent for critical evaluation of fracture risk but also noted inclusion of the feet may be pertinent for whole-body adaptation to physical training by shank analyses. Furthermore, the long-term precision of these methods hinges upon the assumption that the same amount of tissue will be included or excluded for each serial scan as shown in Figure 1. These scan-to-scan discrepancies may lead to large variability in analyzed bone, lean, and fat mass tissues (15); severely impacting precision and a practitioners' ability to accurately assess longitudinal adaptations to training or injury. Another option for practitioners is combining separate regions of interest (ROI) from multiple scans such as isolating the head (12) or the upper body entirely and combining it with included tissue from a cut-off total body scan. However, the subjectivity of cut-off points, which may fluctuate with positioning and over time, may potentially lead to serial assessment variability and ultimately reduce precision. With the increased usage of serial DXA data to inform athlete health, performance, and injury risk, more precise and user-friendly methods are needed to capture the entire body instead of omitting certain body segments.

The purpose of this study was to assess machine precision and the viability of a novel scanning technique for persons taller than 193 cm using the most commonly used DXA models by trained individuals. The research team hypothesized that combining regions of interest from the upper and lower body from 2 separate scans would be a precise way to create a complete total body scan for tall athletes on both GE and Hologic DXA models.

Methods

Experimental Approach to the Problem

A cross-sectional design was used to assess precision of body composition and skeletal measures between 2 total body DXA scans and a third flipped scan. Before the session, subjects were instructed to arrive in athletic clothing free from metal and engage in normal hydration habits. During a single visit, subjects completed screening documentation, provided voluntary written informed consent, were informed of DXA scan procedures, and tested for hydration status and pregnancy status (for females only) before conducting multiple total body scans.

Subjects

Institutional Review Board approval (from Oklahoma State University IRB# 21-497-STW) was obtained, and all procedures were in accordance with the Declaration of Helsinki and its amendments. Subjects' inclusion criteria included men and women between the ages of 18 and 75 years, who weighed less than 114 kg, and were shorter than 193 cm. This ensured that subjects would comfortably fit within the DXA machine's scan area. Exclusion criteria consisted of individuals currently receiving any radiation treatment or women who were currently or planning to become pregnant. A total of 60 subjects aged 34.8 ± 11.9 years (20–66 years old) (male n = 30, female n = 30) completed 3 total body scans. Subjects were overweight

Figure 1. These 2 DXA scans are from a professional basketball player who is 2.11 m tall. Each scan excludes different areas of the body which alters the measures of BMC, adipose, and BFLBM; making serial assessments of body composition inaccurate and potentially misleading.

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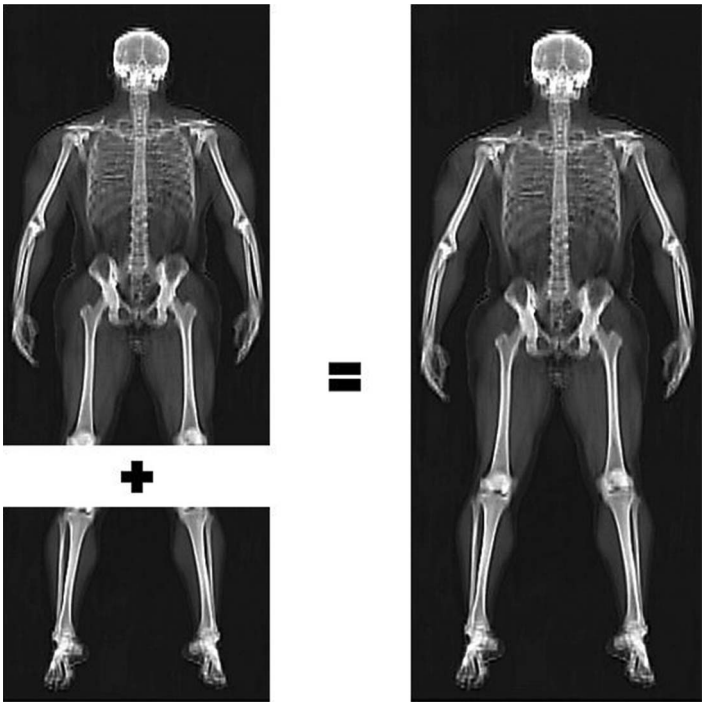

Figure 2. First, using one of the baseline scans a ROI was created from the crest of the head to the distal femoral condyle. Second, another total body scan was completed in the flipped position and a ROI was created from the feet to the proximal tibial plateau. By combining these 2 ROIs a total body scan was recreated.

according to their mean BMI of 26 kg/m^2 (height 1.7 ± 0.1 m, body mass (kg) 75.7 ± 14.8 kg) and self-reported as being White n = 50, Black n = 3, Asian n = 3, or Hispanic n = 4.

Procedures

Dual-energy X-ray Absorptiometry Scans. A GE Lunar Prodigy (n = 15) (LP; enCORE v14.10.022; GE Healthcare, Madison, WI) and a Hologic Horizon A (n = 45) (HA; APEX v5.6.0.5, Hologic Inc, Marlborough, MA) were used for the main analyses, and a GE Lunar iDXA (n = 1) (enCORE v18, GE Healthcare) was used for subsequent exploratory analyses. All subjects had their height (cm) and body mass (kg) measured before providing a urine sample to ensure proper hydration 1.004–1.028 USG by refractometry and confirmation of negative pregnancy status for females. All 60 subjects underwent 3 DXA scans. For the first 2 identical scans, subjects were positioned supine with their hands by their sides and feet were not strapped into a hip block as previously detailed (4,19). Subjects then completed another total body scan in the flipped position proceeding from the feet to the head using the same positioning as before. Machine precision testing for the LP and HA were calculated for bone mineral content (BMC; g), fat mass (FM; g), and bone-free lean body mass (BFLBM; g) using repeated total body scans according to the International Society for Clinical Densitometry (ISCD) guidelines (1). Each scan was performed and analyzed by the same trained technicians.

Region of Interests. It was hypothesized that combining upper- and lower-body region of interest (ROI)s from 2 scans would allow exceedingly tall individuals to have precise serial data of their entire body as shown in Figure 2. This was accomplished by first creating the upper-body ROI from the second baseline scan that proceeded from head to toe under normal manufacture methods. This ROI encompassed all tissues from the crown of the head to the distal femoral condyle. Next a lower-body ROI encompassing all tissue spanning from the proximal tibial plateau to the distal crest of the feet was created from a flipped scan that proceeded from toe to head. This particular anatomical landmark, between the femoral condyle and tibial plateau, was selected for 3 reasons. First, it is comprised of easily identifiable boney landmarks; second, it presents minimal amounts of tissue, reducing the potential for error; and third, these boney landmarks are still present in scans of extremely tall athletes even if their entire skull is included in the scan area. The data from these 2 ROIs were combined and compared with the original total body scan, allowing for the calculation of the CV% for this technique. Dual-energy X-ray absorptiometry data extraction was completed using the automated DXA^2 software to eliminate transcription errors (5).

Statistical Analyses

All statistical procedures were performed using IBM SPSS (v26, Armonk, New York, NY) and $p \leq 0.050$. First, baseline precision testing occurred on both LP and HA DXA machines according to the ISCD standards (1). In summary, 3 consecutive total body scans (2 traditional and 1 flipped) were completed for each subject and compared with one another. The precisions of each machine are expressed as the coefficient of variation (CV%) and were calculated by dividing the population standard deviation by the mean ($CV = (\sigma/\mu) \times 100$). The CV%s of the 2 baseline total body scans and newly created combined ROI scan were also evaluated using paired t-tests, Cohen's d effect sizes (small = 0.2, medium = 0.5, and large = 0.8) (9), and Pearson's correlation coefficients. Finally, Bland-Altman plots with calculation of mean bias and

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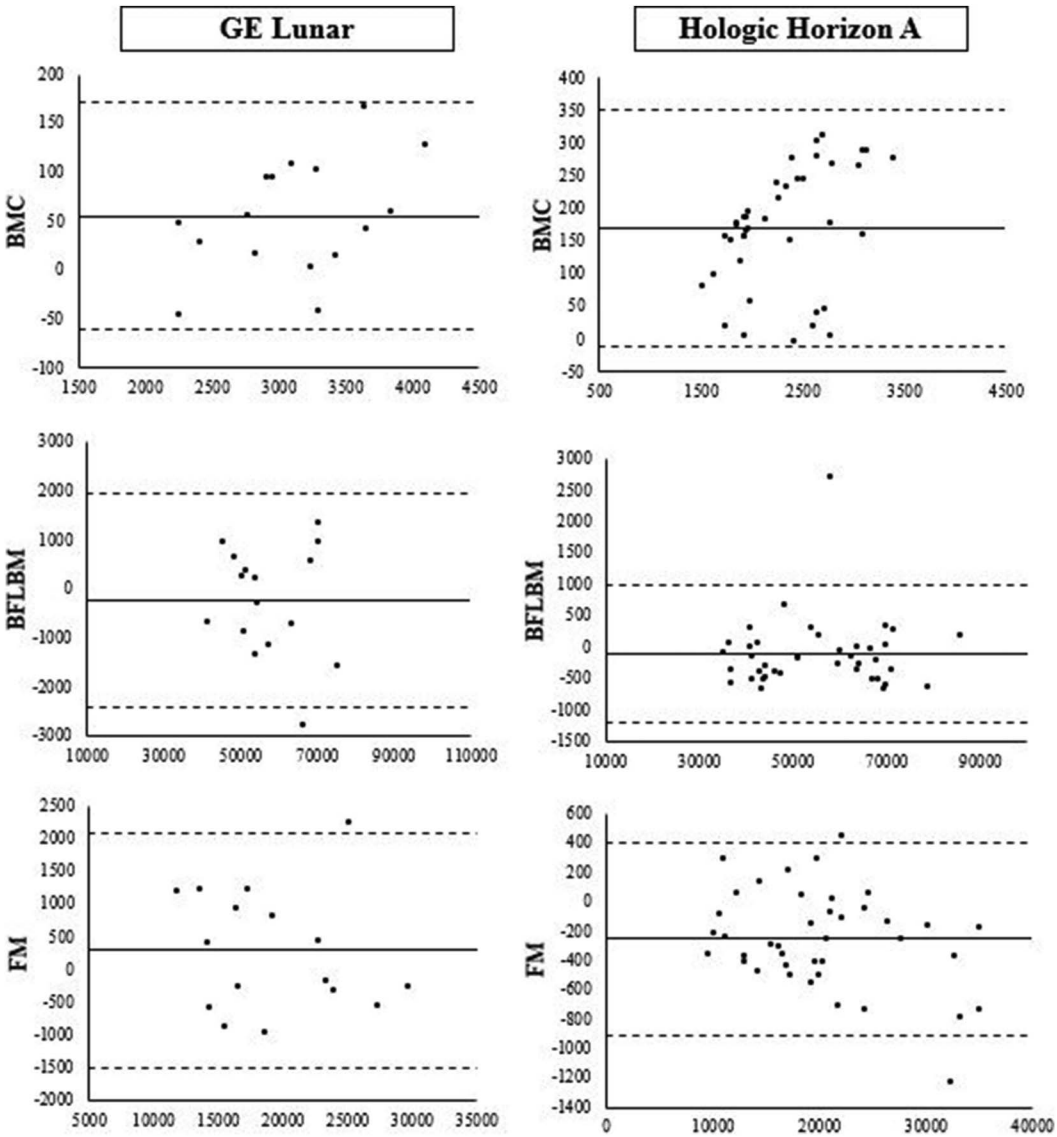

Figure 3. Bland-Altman plots of agreement between the baseline and combined ROI scans for the GE Lunar (left) and Hologic Horizon A (right) models. Dotted horizontal lines represent the upper and lower limits of agreement and the solid horizontal line represents the mean bias between models for BMC, BFLBM, and FM. For both the LP and HA, good agreement for all 3 tissues was found between scans. BMC = bone mineral content; BFLBM = bone-free lean body mass; FM = fat mass.

95% limits of agreement were calculated for BMC, BFLBM, and FM for both DXA models.

Results

For the LP (Table 1), baseline tissue precisions (machine + technician) of BMC and BFLBM were under 1.0%, but FM was over 2.5%. When comparing the initial LP baseline scan with the combined ROI scan, precision worsened by less than 1.0% for BMC, FM, and BFLBM. Paired *t*-tests showed no difference between baseline and combined ROI scans coinciding with trivial Cohen's effect sizes and very strong Pearson's correlation coefficients for both BFLBM ($p = 0.418$; $d = 0.02$; $r = 0.994$, $p < 0.001$) and FM ($p = 0.223$; $d = 0.05$; $r = 0.985$, $p < 0.001$). However, BMC was statistically different between scans ($p = 0.002$); although, the effect size was trivial ($d = 0.10$) and Pearson's correlation coefficient remained very strong ($r = 0.995$; $p < 0.001$). Bland-Altman plots revealed good agreement between scans for BMC (mean bias: 55.58 g, 95% LOA: −60.08, 171.24 g), BFLBM (mean bias: −232.615 g, 95% LOA: −2,420.17, 1954.94 g), and FM (mean bias: 292.99 g, 95% LOA: −1,515.23, 2,101.20 g) within the LP model (Figure 3 left panels).

For the HA (Table 2), all baseline tissue precisions (machine + technician) were under 1.0%. When comparing the initial baseline scan with the combined ROI scan, precision worsened by less than 0.5% for FM and BFLBM, but BMC worsened by over 4%. Paired *t*-tests showed no difference between scans, corresponding with trivial effect sizes and very strong Pearson's correlation coefficients for BFLBM ($p = 0.172$; $d = 0.01$; $r = 0.999$, $p < 0.001$); however, a significant difference was observed with weak effect sizes and very strong correlation coefficients in BMC ($p < 0.001$; $d = 0.00$; $r = 0.984$, $p < 0.001$) and FM ($p < 0.001$; $d = 0.04$; $r = 0.999$, $p < 0.001$); $p < 0.001$). In addition, Bland-Altman plots revealed good agreement between scans for BMC (mean bias: 170.89 g, 95% LOA: −10.58, 352.37 g), BFLBM (mean bias: −106.55 g, 95% LOA: −1,203.21, 990.11 g), and

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Figure 4. Three-step approach for newer Hologic Horizon A and GE iDXA machines. First, ensure all serial scans include the top of the head or other clearly defined boney landmarks like the jaw. Second, on all scans, measure the length of the body included in the scan from top to bottom using the ruler function. Third, using the shortest distance measured across all scans, create a ROI that is that exact height and apply it to all serial past and future scans.

FM (mean bias: −252.33 g, 95% LOA: −905.19, 400.52 g) within the HA model (Figure 3—right panels).

As the BMC was highly skewed on the newer HA, troubleshooting efforts began. Initially, this included deciphering if the +4% error in BMC was from the flipped scan as a whole or just the creation of the upper- and lower-body ROIs. It was found that when the body is scanned in the normal position, from head to toe, custom ROI analyses is highly precise; however, when the feet are scanned first, all bone-specific data are highly inaccurate. After discussions with the manufacturer's engineering team, the head first orientation is required for the tissue calibration algorithms to accurately assess bone. The next question was if this is a manufacture-specific issue or is this a limitation across newer systems in general? The research team repeated the scanning protocol on a single subject using a newer GE Lunar iDXA and found large errors again for BMC (3.71%), BFLBM (4.80%), and FM (7.39%).

Discussion

This investigation sought to compare machine precision and test a novel scanning technique that would allow for accurate and precise serial DXA scan assessments for individuals who exceed the table height limits across different DXA models. Initially, the newer HA model demonstrated tighter precision across all tissues compared with LP, although this could be due in part to the machine's age. The new combined ROI technique added ≤0.63% error across all tissues when using the LP system, providing coaches and practitioners a viable solution to many real-world scanning issues when working with tall athletes. However, this technique added an unacceptable amount of error for BMC on the HA (+4.35%) and iDXA (+3.71%). This still poses the question of what solutions exist for practitioners who have newer GE or Hologic models and routinely scan subjects who exceed the height limits of the DXA table. Below, 3 options will be discussed, which may be beneficial to the coach or practitioner based on the height of the individuals they regularly work with.

The first option is to omit a segment of the body for all serial scans. Previous investigations have debated the efficacy of head or feet exclusion for serial assessment. Nana et al. (17) indicated that exclusion of the head may influence body composition algorithms and skew analysis because of the adipose tissue content of the head being used to create estimates for the rest of the body. Hew-Butler et al. (15) determined that exclusion of the feet may neglect crucial bone information. This is especially pertinent for athlete populations such as basketball who are at elevated fracture risk of that region from their frequent high impact loading profile. Although each technique may be viable for serial assessment, methodological subjectivity may lead to inaccuracies or gaps of crucial data, leading to variability in between-sport comparison and dampening DXAs' diagnostic utility. Therefore, practitioners must take special care to standardize ROI creation and analysis protocols, ensuring the same bone landmarks are used for each subject over time.

The second option is that both GE and Hologic offer extenders that allow the body to be shifted up the table while supporting the head. Although this permits the subject to be more comfortable, if too much of the head is excluded, the same precision inaccuracies such as those observed with the flipped scan may be present. For instance, on the Hologic Horizon A, subjects over 200 cm whose feet are included in the scan may have inaccurate skeletal data because not enough of the head is included for the tissue calibration algorithm to work. Although this accommodation may assist slightly with scan range, analyzed tissue inconsistencies may

Table 1
Precision testing results for the GE Lunar DXA.*†

Tissue	Baseline scan	Combined ROI scan	Precision difference	Paired *t* test (*p*)	Effect size (*d*)	Correlation coefficient (*r*)
BMC (g)	0.97%	1.52%	+0.55%	0.002‡	0.10	0.995‡
BFLBM (g)	0.91%	1.16%	+0.25%	0.418	0.02	0.994‡
FM (g)	2.56%	3.19%	+0.63%	0.223	0.05	0.985‡

*BMC = bone mineral content; BFLBM = bone-free lean body mass; FM = fat mass.
†Data are shown as mean percent coefficient of variation. Paired *t*-tests compared the mean tissue values of the baseline precision scan and the combined ROI scans.
‡Statistical significance $p < 0.050$.

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Table 2

Precision testing results for the Hologic Horizon A DXA.*†

Tissue	Baseline scan	Combined ROI scan	Precision difference	Paired *t* test (*p*)	Effect size (*d*)	Correlation coefficient (*r*)
BMC (g)	0.80%	5.15%	+4.35%	<0.001‡	<0.01	0.984‡
BFLBM (g)	0.48%	0.48%	<0.01%	0.172	0.01	0.999‡
FM (g)	0.95%	1.26%	+0.31%	<0.001‡	0.04	0.999‡

*BMC = bone mineral content; BFLBM = bone-free lean body mass; FM = fat mass.

†Data are shown as mean percent coefficient of variation. Paired *t*-tests compared the mean tissue values of the baseline precision scan and the combined ROI scans.

‡Statistical significance $p < 0.050$.

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not warrant this solution for serial assessment of athletes who are exceedingly tall.

A third option is the use of the shortest applicable ROI (Figure 4). For retrospective analyses of scans, the coach/practitioner first must assess all old scans for a consistent boney landmark such as the top of the skull, jaw, or bottom of the feet then using the ruler function, find the scan with the shortest distance. This measurement coincides with the least amount of the body included in a scan. Next, they must create a ROI that is that shortest distance and apply to all retrospective scans to ensure the same amount of tissue is being assessed from scan to scan. This 3-step approach ensures the same amount of the body is included in each of the past scans and can be applied to all future scans allowing practitioners to have more confidence in tissue changes over time. A key consideration for this method will be an athletes' skeletal age. If the athlete is still gaining height, such as new professional basketball players who often enter a league between 18 and 22 years of age, the shortest ROI method will begin to exclude more and more tissue, reducing the precision of the serial assessments.

This study assessed machine precision and introduced a novel scan technique for LP and HA DXA models that may reduce tissue analysis inconsistencies through creation of custom ROIs with easily identifiable and consistent cutoff points. The flipped method was successful on the LP, providing users a precise and easy method to improve data collection with tall athletes. This scanning technique was also partially precise for newer DXA models; however, bone errors were high. In the future, DXA manufactures will need to create a larger table option or modify software calibration algorithms allowing practitioners working with tall athletes to precisely assess skeletal health and body composition adaptations over time.

Practical Applications

The use of DXA systems in athletics is only gaining popularity; however, athletes who are taller than 193 cm often have tissue omitted across multiple time points, reducing the data's utility. There are 3 main considerations: First, this is one of the few studies to evaluate a practitioner-friendly scanning technique that guides users through ROI creation to allow for precise data collection in tall athletes over time. Second, coaches, staff, and practitioners with the HA or iDXA must carefully consider the limitations discussed here when creating their team's standard scanning procedures for tall athletes. Finally, DXA users need to be able to discuss their athlete's data over time within the context of their specific machine's precision. If the DXA being used has not had baseline precision testing, then serial assessments are suspect to inaccurate and potentially dangerous interpretations.

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